# Exact Evaluation of Time-Series Anomaly Metrics by Local Event Statistics

Youngmin Ko
rhdudals0505@naver.com
Independent Researcher
Republic of Korea

## ABSTRACT

Time-series anomaly evaluation repeatedly scans stored detector profiles at many thresholds. We derive local event statistics for two existing operators, preserving their numerical targets. For dynamic VUS, inactive precision–recall events vanish while ROC needs a predecessor correction, yielding a support-sensitive exact representation. For range precision and recall, coupled overlap and fragmentation updates maintain the entire trajectory as predicted intervals merge. A finite-feature weight theorem accounts explicitly for cardinality evaluation; the canonical complete trajectory takes $O(n \log n)$ arithmetic operations and is comparison-optimal for query-associated outputs. We separate classical sorting, component trees and caching from these metric-specific identities, and compare against same-compiler raster, closed-form interval and conventional support controls. The original 73-series study is extended with 26 separately selected series and unmodified public detectors. Exhaustive weighted curves improve over both compiled controls at the aggregate level in three collections on two hosts. A native TimeEval integration with flat, cardinality-one range PR-AUC, including score reads, all three requested metrics and result writing, improves by 1.93–2.76 times over compiled raster at all native observed queries across four Apple/Linux study executions. An additional interval control retains the gain; a separately frozen five-session affinity study gives a 1.92-fold exhaustive gain. Low-cardinality losses, small selection effects on the separate panel, small or unsupported default-consumer gains and a 1.031-fold VUS-only nine-metric consumer gain delimit the practical benefit.

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The source code, data, and/or other artifacts have been made available at https://github.com/mandu5/sparse-vus/tree/448ad1b714a5daa13feeba617b10e02efef674fe.

## 1 INTRODUCTION

Time-series anomaly benchmarks repeatedly join stored detector scores with labels to evaluate and compare models. A range-aware metric can require a complete pass through the same series at every score threshold, and a volume metric repeats this work across temporal widths. Sampling the thresholds reduces work but changes the numerical query. Our question is which local statistics suffice to answer the original query exactly.

We study two established operators. Dynamic Volume Under the Surface (VUS) integrates precision–recall (PR) or receiver-operating-characteristic (ROC) curves over temporal widths [2, 18]. Tatbul range precision and recall reward positional overlap and penalize fragmentation of true and predicted intervals [24]. Both are used in public evaluation software. TimeEval's maintainers explicitly identify slow range metrics and threshold sampling as an optimization opportunity [26]. These operators have different state dependencies: VUS can omit inactive support events, whereas range precision changes even when a point outside every true interval joins two predicted components.

Consider the four-point example $y = (1, 0, 1, 0)$ and $s = (3, 2, 1, 0)$. Removing off-label events and connecting the remaining ROC endpoints changes zero-width VUS-ROC from .625 to .8125. An omitted run must instead contribute its predecessor rectangle. For range metrics, dropping an off-label point can change predicted intervals and therefore their precision denominators. The same intuitive "ignore irrelevant points" shortcut cannot serve both operators. We derive the sufficient state for each.

For VUS, complete global ties, predecessor rectangles and two-distance interval geometry give a support-sensitive representation. For range metrics, endpoint activations update one recall contribution and at most three precision contributions. Fixed-label moment prefixes support moving positional weights, including a middle bias whose interval midpoint changes after a merge. A finite-feature theorem covers endpoint-dependent piecewise weights; the implementation provides four canonical biases and three cardinality factors. Known query grids can use ordinary offline bucketing instead of sorting every score. The new result is the metric-specific coupled state, rather than sorting, prefix sums, caching, or component trees.

The data-management setting is repeated analytics over stored score columns. We provide complete-call and read-to-result-write experiments, including a native public TimeEval consumer, and compare with same-compiler conventional controls. We do not propose a detector, a new accuracy metric, an unlabeled model selector, or a database engine. Our earlier public Sparse VUS 0.3.0 release [9] is an unrefereed software precursor; its speed is separated from subsequent shared-label computation. It contains no weighted range trajectory implementation.

The contributions are: (i) exact support-event identities for dynamic VUS and a general local-update theorem for weighted range

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trajectories; (ii) algorithms and ablations separating these identities from compilation and work sharing; and (iii) source-bound evaluation on public data, including a separately selected panel with unchanged public detectors and native consumer execution. We retain reused recordings, low-cardinality losses, non-beneficial alternatives, and consumers whose other metrics dominate total cost.

## 2 TARGET AND EXACT EVENT REPRESENTATION

For VUS, let $y \in \{0,1\}^n$ contain both classes and $s \in \mathbb{R}^n$ contain finite scores after binary64 conversion. Larger scores indicate greater anomaly evidence. Let $P_0 = \sum_i y_i$, and let the $K$ maximal original anomaly intervals have inclusive endpoints $[a_k, b_k]$ in temporal order. Widths are the integers $w = 0, \dots, L$. The continuous label $a_i^{(w)}$ is one on original positives. An interval contributes $\sqrt{1 - d/w}$ outside its endpoints at integer distance $1 \le d \le \lfloor w/2 \rfloor$; contributions are summed and clipped at one. At $w = 0$ no ramp is added. Expanded intervals that overlap merge, while adjacent nonoverlapping intervals remain separate. This is the original dynamic VUS target pinned to the official `RangeAUC_volume_opt` source at revision `467bace0`, configured for exhaustive thresholds [25]. It is not shared by all software named VUS.

For a threshold $t$, define

$$p_i(t) = \mathbf{1}\{s_i \ge t\}, \qquad C(t) = \sum_i p_i(t), \tag{1}$$

$$A(t) = \sum_i y_i p_i(t), \qquad B(t) = \sum_i (1 - y_i) a_i^{(w)} p_i(t), \tag{2}$$

$$T(t) = A(t) + B(t), \qquad P(t) = P_0 + B(t)/2, \tag{3}$$

$$N(t) = n - P(t). \tag{4}$$

Let $H(t)$ count expanded components hit by at least one predicted positive, and let $K_w$ count those components. The curve coordinates are

$$R(t) = \min\{T(t)/P(t), 1\}\, H(t)/K_w, \tag{5}$$

$$F(t) = (C(t) - T(t))/N(t), \qquad \pi(t) = T(t)/C(t). \tag{6}$$

Precision is used only at nonempty retained thresholds, where $C(t) > 0$; the initial origin requires no precision value. The denominator is positive: $P(t) \le (n + P_0)/2 < n$. ROC integrates trapezoids from the origin through complete distinct descending score thresholds to the explicit terminal $(1, 1)$. PR uses the right-step sum $\sum_j (R_j - R_{j-1})\pi_j$, with $R_0 = 0$. Each VUS scalar is the arithmetic mean over every width. Signed horizontal ROC increments are preserved; the derivation does not replace a dynamic denominator by a fixed one.

Let $S_w = \{i : a_i^{(w)} > 0\}$, $Q_w = |S_w|$, and $Q = Q_L$. For an active event with score $v_j$, precompute the complete global counts $c_j^{>} = |\{i : s_i > v_j\}|$ and $c_j^{\ge} = |\{i : s_i \ge v_j\}|$. They include positions outside $S_w$ and outside $S_L$.

Lemma 1 (Inactive threshold runs). *Between consecutive active score events at a fixed width, $T, B, P, N, H, R$ are constant. Omitted thresholds contribute a horizontal ROC rectangle and zero PR area. At an active score, its complete global numerical tie is a single transition.*

Proof. A newly admitted inactive point has zero weight and cannot hit an expanded component. Only $C$ changes. The horizontal increments $(\Delta C)/N$ telescope at fixed height $R$. All positions tied with an active event enter at its global threshold, so both counts must retain them. □

If the previous retained state is $(T_o, N_o, R_o, c_o)$, event $j$ contributes

$$f_j^{-} = (c_j^{>} - T_o)/N_o, \tag{7}$$

$$\Delta\mathrm{ROC}_j = (c_j^{>} - c_o)R_o/N_o + (F_j - f_j^{-})(R_j + R_o)/2, \tag{8}$$

$$\Delta\mathrm{PR}_j = (R_j - R_o)T_j/c_j^{\ge}. \tag{9}$$

The initial state is $(0, n - P_0, 0, 0)$, and afterwards $c_o = c_j^{\ge}$. Connecting retained endpoints without the intervening rectangle is incorrect. For example, $y = (1, 0, 1, 0)$, $s = (3, 2, 1, 0)$ and $L = 0$ give ROC area 0.625; connecting support endpoints alone gives 0.8125.

Lemma 2 (Two-distance geometry). *The weight of a non-original point is determined by its distances $d_1, d_2$ to the nearest two distinct original intervals. It is zero if $d_1 > \lfloor w/2 \rfloor$, one if $d_2 \le \lfloor w/2 \rfloor$, and $\sqrt{1 - d_1/w}$ otherwise.*

Proof. Every active ramp has weight at least $1/\sqrt{2}$. Two contributions therefore clip to one. If exactly one interval contributes, its distance determines the weight. Distances to ordered intervals are monotone moving away on each side, so two neighbors on each side contain the nearest two intervals. □

Theorem 1 (Support-event evaluation). *In a unit-cost model with score comparisons and metric arithmetic, including square roots, both exhaustive scalar integrals can be computed in $O(n \log(Q+1) + (L+1)Q)$ time and $O(n + Q)$ auxiliary words.*

Proof. Extract original intervals and maximum support in $O(n)$ time. Sort the $Q$ support scores and group equal values. Binary-search all $n$ scores among these groups to obtain complete global counts $c^{>}, c^{\ge}$ without sorting off-support scores, at cost $O(n \log(Q + 1))$. The ordered-interval geometry can be prepared within this bound. At each width, compute support weights, component membership, grouped masses and first component-hit events in $O(Q + K)$ time. A scan of the at most $Q$ event groups integrates the preceding rectangles, complete tied transitions and terminal ROC tail. The inactive-run lemma makes this identical to the exhaustive target in exact metric arithmetic. Stream the width mean using input-, support- and interval-sized arrays. Since $K \le Q \le n$, the stated bounds follow. □

Proposition 1 (Fixed-binary64 preparation). *For finite binary64 scores, complete global tie counts, the score ordering of maximum support, and its two-distance geometry can be prepared in $O(n + Q + K)$ time and $O(n + Q)$ auxiliary words. Discrete work uses a word-RAM with input indices, counts, widths, and intermediate index arithmetic fitting words. Metric arithmetic, including square roots, has unit cost.*

Proof. Normalize signed zeros. Complement a negative score's bit pattern and flip the sign bit of a nonnegative score, then complement the result to obtain a descending numerical key. Eight stable byte-counting passes order these 64-bit keys in $O(n)$ time. Scan complete ties and retain their starting and ending global ranks. Filter the permutation to maximum-support positions. A temporal scan with a monotone interval pointer computes nearest distances and owners in $O(Q + K)$ time. All arrays are input- or support-sized. □

Radix sorting and floating-point key transforms are established techniques [4, 7]. This assumption does not give linear sorting of arbitrary real scores or a bit-complexity bound for growing precision. Conversion defines numerical ties, including collisions between distinct large integer inputs.

The same support-event width kernel combined with this preparation gives $O(n + (L + 1)Q)$ time. This is the basic Sparse VUS evaluator; its improvement over a dense rank sweep depends on $Q$, rather than recall saturation. The released PR-only implementation retains $L + 1$ scalar width results, adding $O(L)$ words; streaming removes this storage. Input-sized ranking arrays remain necessary in this construction, so the memory is not support-only.

## 3 PRIOR VUS ALGORITHMS AND SHARED LABELS

Boniol et al. already distinguish maximum-buffer support from unaffected regions and reuse static work [2]. Klein's fast-VUS uses support-score cumulative sums; Benchopt uses dense cumulative sums and segment maxima [1, 8]. Their public exact configurations use $H = n$, rather than the sampled defaults. The inspected Klein implementation still constructs full-length coverage arrays at each width; Benchopt builds full-series cumulative sums and returns both ROC and PR. These array operations are distinct from the support-event representation. We compare equal returned work separately from the cost of an available PR-only API.

Sylligardos's efficient VUS work and the public FF-VUS implementation are important acceleration antecedents [22, 23]. The latter uses linear/max ramps differing from our pinned dynamic target, rounds scores in its wrapper, and has an approximate existence fallback in its Torch path. We retain source-level semantic witnesses; native FF-VUS timings would not be equal-output speed ratios. No experiment here establishes superiority over an unmeasured GPU evaluator. Complete ties and trapezoidal sweeps are established ROC techniques [5]; full-recall PR stopping is present in scikit-learn 1.0.2 [21]. Neither is claimed as new.

For $M$ profiles sharing labels, the *geometry* variant prepares original intervals, support and distances once. The *stream* variant additionally interchanges width and model loops: each width's weights and component identifiers are prepared once, while all model-specific grouped masses and complete events remain separate. Under Proposition 1, this costs $O(Mn + (L + 1)MQ)$ time. Its retained workspace after conversion and validation is $O(MQ + n + Q + K)$: two model-by-support rank/count arrays occupy $16MQ$ bytes. The public NumPy validation and conversion can allocate $M \times n$ arrays, giving the more conservative peak bound $O(Mn + MQ + n + Q + K)$. Sharing can therefore use more memory. Caching and loop interchange are standard optimizations; their increment is measured against the released evaluator.

An earlier active-prefix alternative uses reversible interval updates and complete-tie recall saturation, with predicate $A + B/2 \geq P_0$ and $H = K_w$. Its conditional bound and a single-interval run variant remain in the artifact. The default-width prefix route did not pass its runtime target. This outcome motivates measuring actual visited work rather than assuming early saturation.

## 4 INCREMENTAL WEIGHTED RANGE PRECISION AND RECALL

Tatbul range metrics score intervals using positional overlap and fragmentation [24]. TimeSeAD already evaluates every observed score threshold [3, 28], and TimeEval exposes weighted range PR curves [27]. We address their repeated recomputation by maintaining both sides of the metric as predicted intervals appear, extend, and merge.

*Target.* Let $T_j = [a_j, b_j]$, $j = 1, \ldots, K$, be the maximal positive intervals of fixed binary labels $y$. Predictions use $p_i(t) = \mathbf{1}\{s_i > t\}$, distinct from the VUS convention. Let $C$ range over current maximal predicted intervals. For $X \in \{P, R\}$, weights $w_X(i; a, b) \geq 0$ depend on the scored interval's endpoints, with $Z_X(a, b) = \sum_{i=a}^{b} w_X(i; a, b) > 0$. Define

$$W_j = \sum_{i \in T_j} p_i w_R(i; a_j, b_j), \qquad N(C) = \sum_{i \in C} y_i w_P(i; a, b).$$

Write $c_j$ for the number of predicted intervals intersecting $T_j$ and $h(C)$ for the number of true intervals intersecting $C$. For $\alpha \in [0, 1]$,

$$r_j = \alpha \mathbf{1}\{c_j > 0\} + (1 - \alpha) g_R(c_j, |T_j|) \frac{W_j}{Z_R(a_j, b_j)}, \tag{10}$$

$$\psi(C) = g_P(h(C), |C|) \frac{N(C)}{Z_P(a, b)}, \tag{11}$$

$$R = \frac{1}{K} \sum_j r_j, \qquad P_\rho = \frac{\sum_C \rho(C) \psi(C)}{\sum_C \rho(C)}. \tag{12}$$

Zero-overlap terms and empty averages are zero; factors are never evaluated at count zero. Effective cardinality factors $g_P, g_R \in [0, 1]$ are pure functions of count, interval length, and fixed configuration. Let $C_\gamma(n) \geq 1$ bound either factor's arithmetic/comparison evaluation cost for $1 \leq c \leq L \leq n$, using $O(1)$ auxiliary arithmetic words. Canonical factors satisfy $g(1, L) = 1$. The positive, endpoint-local aggregation weight $\rho$ costs constant time: $\rho = 1$ gives macro precision and $\rho = |C|$ gives length-weighted precision over *all* predicted intervals.

*Weight family.* For every $[a, b]$, suppose each weight partitions the interval into at most $q \geq 1$ contiguous integer pieces $I_r(a, b)$, on which

$$w_X(i; a, b) = \sum_{k=0}^{d-1} \theta_{Xrk}(a, b) \phi_k(i), \qquad d \geq 1.$$

Features are endpoint-independent and their values and prefixes require $O(dn)$ preprocessing. Piece endpoints and coefficients require $O(qd)$ work per scored interval. These are explicit arithmetic-cost assumptions, not permission for arbitrary root

finding or mask-dependent callbacks. Piecewise polynomials of degree at most $d-1$ use $\phi_k(i) = i^k$. Fixed-label prefixes of $y_i\phi_k(i)$ and $\phi_k(i)$ recover $N(C)$ and $Z_P(C)$ by summing $qd$ coefficient-weighted prefix differences. Recall weights have fixed endpoints and can be precomputed once.

Theorem 2 (Joint weighted range trajectory). *Under these assumptions, $H$ arbitrary threshold queries on $n$ points take*

$$O\big(n\log(n+1) + H\log(H+1) + n(qd + C_\gamma(n)) + H\big)$$

*time and $O(dn+n+H+qd)$ words in a unit-cost model of exact arithmetic and comparisons. Constant-cost factors recover $O(n\log(n+1)+H\log(H+1)+qdn+H)$. The stated coefficient and cardinality evaluation costs are assumptions; the bound counts arithmetic operations, not exact-rational bit complexity.*

Proof. Maintain the active mask, every $(W_j, c_j)$, and the sums defining $R$ and $P_\rho$. Precompute in $O(n)$ time an owner array mapping each labeled point to its true interval. For each active component $[a, b]$, store its left boundary at $b$ and its right boundary at $a$; interior entries may be stale. Initially all statistics are zero.

Inserting $i \in T_j$ adds its fixed weight to $W_j$ and changes

$$c_j \leftarrow c_j + 1 - \mathbf{1}\{i-1 \text{ active in } T_j\} - \mathbf{1}\{i+1 \text{ active in } T_j\}.$$

Replace only $r_j$. Restricted to a contiguous true interval, active runs correspond exactly to intersecting predicted components. An insertion outside all true intervals cannot connect two fragments inside one such interval, because their unique path lies inside it. Existence uses $c_j > 0$, even when a detected point has zero weight.

For precision, remove the zero, one, or two adjacent component contributions and add their replacement interval. An active neighbor of an inactive point must be an endpoint, so the stored boundaries identify both components in constant time; stale interior entries are never queried. Update the new outer endpoints and both precision sums. With $\sigma_i$ indicating a true interval start and $D(t) = \sum_{i<t}\sigma_i$,

$$h([a,b]) = D(b+1) - D(a) + y_a - \sigma_a.$$

Thus each insertion evaluates at most three precision and two recall terms in $O(qd + C_\gamma(n))$ work and preserves every invariant. Fixed-weight preprocessing costs at most $O(qdn)$ using one coefficient list as scratch. Sort scores and queries descending, activate all points strictly above each query, and emit only after complete ties. Each point activates once; output restoration costs $O(H)$. The prefix tables, maintained state, scratch, and outputs give the space bound. □

Corollary 1 (Comparison cost of the complete trajectory). *For constant $q, d$, $C_\gamma(n) = O(\log(n+1))$ and $H = n$, the trajectory takes $O(n\log(n+1))$ operations. This is worst-case comparison-optimal for arbitrary distinct real scores and query-associated exact outputs, already with flat weights and cardinality one.*

Proof. Use $n-1$ consecutive positive labels with arbitrary distinct positive scores, followed by one negative label of score zero; take flat weights, cardinality one and $\alpha = 0$. Request $t_i = s_i$ in input order and write $J$ for the positive positions. For $i \in J$, $R(t_i) = |\{j \in J : s_j > s_i\}|/(n-1)$ reveals its rank. A comparison-based algorithm distinguishes $(n-1)!$ orders and needs $\Omega(n\log n)$ comparisons. This is the standard sorting reduction. An anonymous multiset of recalls would not identify ranks. The result concerns full query-associated curves, not just their maximum, unrestricted algebraic operations, or fixed-width radix algorithms. □

*Constructive relation to a component tree.* A one-mask Tatbul evaluator can traverse ordered true and predicted intervals in linear time and compute each positional overlap in closed form; it need not enumerate every pair of intervals [24]. Our fused raster avoids all-pairs overlap enumeration but still evaluates point weights. The separate conventional interval control uses closed-form overlap and fixed-label prefixes at each query, without cross-threshold state. A classical threshold component tree additionally supplies births, extensions, merges, endpoints and component sizes across thresholds [14, 16]. Size-based precision/recall through such a tree also predates this work [20]. These quantities alone do not specify the Tatbul trajectory: identical scores with different labels give the same tree but different overlap and fragmentation statistics.

The required augmentation is explicit. Precision uses the true-start prefix $D$, label-weight prefixes, and the two global sums $\sum_C \rho(C)\psi(C)$ and $\sum_C \rho(C)$. Recall additionally needs one $(W_j, c_j)$ pair per true interval and $\sum_j r_j$. At each activation:

> Remove the precision terms of adjacent active components.
> If $i \in T_j$, replace $r_j$ using the updated $W_j, c_j$.
> Join the adjacent components through $i$; update the outer endpoints.
> Compute the replacement component's $h, N, Z, \rho$ from the prefixes and add its precision term.
> After the complete eligible tie or query bucket, emit the two global averages.

An explicit component tree augmented with these identities is an alternative implementation of the same algorithmic class. The endpoint representation avoids materializing a hierarchy that these queries do not consume; we claim no asymptotic superiority over that augmented tree. The metric-specific contribution is the coupled sufficient state and its local identities.

Moving weights require more than a single component count and moment. On positions $0, 1, 2, 3$, label patterns 1001 and 0110 both have count two and first moment three, yet middle-weight masses two and four. Prefix queries on the two sides of the current midpoint recover this distinction after every merge without revisiting the component's points.

An off-label bridge can change precision even though every recall contribution is unchanged. With $y = (1, 1, 0, 0, 1, 1)$ and $p = (0, 1, 1, 0, 1, 1)$, reciprocal cardinality and middle precision bias give macro precision $(1/2 + 1)/2 = 3/4$. Activating zero-based index 3 joins $[1, 2]$ and $[4, 5]$ into $[1, 5]$: its weight mass is 4 of 9 and it intersects two true intervals, so precision becomes $(1/2)(4/9) = 2/9$. The fixed true intervals retain their overlap and fragmentation. This is why off-label activations and moving normalizers must both participate in the precision update.

*Finite query grids and consumers.* Full score sorting is unnecessary for a known grid. Binary-search each score's first strictly eligible query among descending thresholds, then counting-bucket

its activation. Processing complete buckets preserves the queried masks regardless of internal order, giving

$$O\big(n\log(H+1)+H\log(H+1)+n(qd+C_\gamma(n))+H\big).$$

Already ordered queries omit their sorting cost. A consumer that subsequently sorts by recall, such as TimeEval, adds $O(H\log(H+1))$; its endpoints and integration convention remain part of its contract. Range recall need not be monotone. Strict observed-score queries exclude the all-active state below the minimum; include a lower sentinel when required.

*Implemented family.* For $u = i - a + 1$ and $L = b - a + 1$, the canonical choices are

| bias | $w(u, L)$ | $Z(L)$ |
|---|---|---|
| flat | 1 | $L$ |
| front | $L + 1 - u$ | $L(L+1)/2$ |
| back | $u$ | $L(L+1)/2$ |
| middle | $\min(u, L + 1 - u)$ | $\lfloor (L+1)^2/4 \rfloor$ |

They need at most two pieces and first moments. Middle precision queries both sides of the moving midpoint; an unsplit count and moment are insufficient. Precision and recall biases are independently selectable. Implemented cardinalities are 1, $1/c$, and TimeSeAD's $((L-1)/L)^{c-1}$, with value one at $c = 1$ [28]. The first two require $O(1)$ exact arithmetic operations; repeated squaring evaluates the third in $O(\log(n+1))$ operations with $O(1)$ arithmetic registers. Thus all three retain the complete-trajectory bound, whereas the grid's linear update term requires constant-cost factors or fixed-format evaluation. One shared cardinality choice and macro or length-weighted precision are implemented. Separate generic callbacks, polynomial weights, and the broader finite-feature interface are theorem-only generalizations.

*Numerical scope and antecedents.* Integer counts and first moments avoid cancellation from subtracting rounded floating prefixes. For the canonical family, $n \le 2^{31} - 1$ bounds all intermediate products within signed int64; this restriction does not certify higher-degree polynomials. The improved factor uses a count-one branch and $\exp((c-1)\log1p(-1/L))$. Division and accumulated differences still round: real-arithmetic equality is neither bitwise agreement nor a length-independent floating-point certificate. Scores are finite binary64; query infinities provide sentinels, and ties are complete.

Tatbul already gives closed-form overlap evaluation [24]. Component trees and attribute maintenance are classical [16]; Menotti et al. give linear-time 1D construction with height, surface, and volume attributes [14]. We claim neither new positional biases nor a new component tree. The contribution is the coupled, label-conditioned overlap and fragmentation state across thresholds; ordering and bucketing are standard offline techniques.

## 5 EXPERIMENTAL DESIGN

We distinguish the original VUS/range panel, a separately selected weighted panel, and native consumer queries. Each follow-up freezes its own methods, inputs, orders and censoring rule before its measurements; it is not a retrospective enlargement of the original protocol. All exact comparisons hold the metric definition and query set fixed. Comparing sampled and complete queries is instead an accuracy–cost analysis. The study-to-output map is: dynamic VUS returns integrated ROC/PR values (primary VUS-PR comparison in Table 1); flat and weighted range studies return strict-query precision/recall/F1 trajectories and maximum F1 (Tables 2, 3, 6); native TimeEval returns three integrated metrics, with inclusive range queries and synthetic endpoints (Tables 4, 5 and 7; process-session follow-up in Table 8). These contracts are compared internally, not treated as interchangeable outputs. An initial weighted Mac run was stopped after 12 WSD cases and one Stock query contract. Before a complete rerun, we separated paired-order timing uncertainty from recording sensitivity and bound the independent auditor in the protocol. Cohort, profiles, semantics, methods, orders and hypotheses were unchanged. The 5,400 partial records are retained but excluded from the final study; its analysis protocol was therefore fixed after this partial pilot, not before every observation in the research process.

*Original panel and its dependence.* The pinned TSB-AD-U archive [12] supplies all eligible NAB, SMAP and SMD recordings after six named development exclusions, source-labelled synthetic exclusions and exact observation-plus-label deduplication. The 73 recordings (20/17/36 by collection) have lengths 1,000–28,696. Eight fixed CPU procedures produce Isolation Forest, LOF, nearest-neighbor, one-class SVM, PCA, KMeans, HBOS and shrinkage-Mahalanobis scores. They use trailing length-16 windows, an evenly spaced unlabeled fit set capped at 2,048 points, and seed 20261024. These authored offline transductive procedures are workload generators, not reproductions of eight tuned benchmark pipelines. LOF novelty-mode scoring includes fit points. All 584 outputs retain binary64 scores without jitter or rounding; distinct counts range from 3 to 28,696. A five-distinct-cluster warning for requested 16-cluster KMeans remains in the record.

Sixty-two recordings were used by name in older repository studies. SMD's 36 cases have nine distinct label sequences; 18 share labels with excluded development recordings. We report label-block weighting and exclusion sensitivity. These checks do not establish independent physical sources, and fresh scores do not make the recordings unseen.

*Separately selected public-detector panel.* Before fitting or metric outcomes, we screen archive candidates against 563 prior provenance entries by name, exact labels, exact observations, and z-normalized observations rounded to ten decimals. We select 12 WSD recordings at fixed length quantiles among 91 eligible cases, all 12 eligible Stock cases, and both eligible MSL cases. The 26 recordings contain 386,447 observations. This screening removes the declared forms of reuse; it cannot prove absence of shifted variants or common physical sources. In particular, MSL is only two recordings, not a broad new-domain sample.

Unchanged public FFT and COPOD APIs at TSB-AD revision `ed74348e` generate two profiles per recording, using the full univariate input and their upstream preprocessing. We perform no extra quantization or model selection. The fixed seed is 20261024, one thread is used, and the per-fit limit is 120 seconds. All 52 fits complete, with no replacement or retry. A path-packaging error occurred after the original Mac fits; the receipt was recovered from their immutable outputs with zero repeated fits. Original producer

and recovery sources are archived. The Linux replay uses the corrected packaging path and regenerates every score from raw inputs. Following a failed input-identity check, a separate decoding contract was frozen before fresh Linux fits. It restores one original Mac binary64 observation that the Linux parsers round one ULP differently. Its raw-file digest, decimal token, row index, adjacent binary64 values, and full before/after array hashes are explicit. All original input-identity checks remain mandatory; this reproduces the frozen typed input rather than asserting cross-platform identity of the original parser.

*Comparators and contracts.* VUS compares stream, geometry-only, released sparse radix, active prefix, matched dense radix, unchanged Klein, and a separately frozen Benchopt pair. An author-created conventional support control shares validation, radix ordering and global counts, but uses direct per-width ramps instead of nearest geometry. A minimal Klein ablation changes only its threshold list to distinct maximum-support scores. These controls isolate compiler, support and sharing effects; neither is an unchanged public release.

The original flat/reciprocal range study compares vectorized prefix/interval recomputation, fused compiled raster, score-sorted sweep, and query buckets at 100 evenly spaced strict thresholds. The exhaustive contract evaluates every observed score under $s_i > t$. It excludes the all-active state below the minimum. The weighted follow-up uses raster, sweep and bucket with all 12 combinations of four equal precision/recall biases and three cardinality factors, macro precision, and recall $\alpha = .2$. Mixed biases and length-weighted precision are correctness cases, not extra timed workloads. The same compiler and arithmetic formulas are used for the raster and incremental methods.

A subsequent comparator audit identified an additional conventional control: Tatbul's closed-form ordered-interval traversal. In a separate frozen ablation, we precompute fixed-label count/moment prefixes, discover all predicted intervals by a complete score scan at each query, and compute recall from ordered intersections and closed-form positional sums. No metric state is reused across thresholds. Six permutations remeasure raster, this interval control, and the incremental backend (bucket for grid100, sweep for distinct) on the unchanged full panel and all 12 semantic choices. This prospective follow-up isolates conventional per-query optimization from joint state maintenance; it does not replace the original frozen raster results.

*Native TimeEval consumer.* We pin the complete TimeEval source at `7c362275`, instantiate its unchanged `MetricComputor`, and request ROC-AUC, PR-AUC and range PR-AUC. The range configuration is `DefaultMetrics.RANGE_PR_AUC`, whose parameters differ from the range-curve class constructor defaults. The native loader reads dataset labels and stored scores, computes all three metrics and writes its result CSV. Only the inner range curve callback is replaced. Validation, score sanitization, constants, threshold selection, synthetic endpoints, recall/precision ordering and trapezoidal integration remain native. This experiment uses flat bias, cardinality one, and zero existence reward, matching the native default. Its $\geq$ query is mapped to the strict kernel using the preceding representable binary64 threshold. It is a different query contract from the strict observed-score F1 study.

We compare the unchanged public callback, compiled raster, and buckets in two contracts: native nominal `max_samples=50`, and every nonminimum observed-score query, retaining the native synthetic endpoints. Nominal 50 is not a hard cap: the Mac inputs produce 2–90 actual queries per profile. The consumer preserves that behavior. Batches are all 52 profiles together and each of the three collections separately. Six method permutations balance order, giving 144 scheduled trials. A workflow is censored at 600 seconds; after its first censor, the five remaining same-method/contract/ batch repetitions are explicitly not run. Censored calls provide no completed public-output parity claim or ordinary timing confidence interval.

*Timing and uncertainty.* Warm operator calls include validation, preparation, sorting, allocation, curve production and numerical model selection. Whole-consumer clocks include CSV loading, all three metrics and result-CSV write/close; they exclude imports, JIT warm-up, trial-directory preparation and audits. Detector fitting, cold storage and `fsync` are outside the claimed workload. A sum of case medians is not described as directly timed collection latency. Numerical libraries use one thread, but shared-host load is not controlled.

Original VUS case medians use 12 balanced orders, range-grid medians eight, and the original exhaustive range comparison two AB/BA calls per method. Paired case log ratios are bootstrapped 10,000 times, with a three-collection Bonferroni 95% interval per comparator. Label-block sensitivity addresses repeated SMD geometry. These are fixed-panel sensitivity analyses, not estimates from an independent random population of recordings.

For the new weighted panel, each case/contract/semantics combination uses six method permutations. Primary uncertainty resamples paired order indices within each recording, sharing indices across both methods and all 12 semantics, recomputes medians, then averages log ratios equally across semantics and recordings. It is conditional timing uncertainty on the fixed panel. A separate recording-reweighting bootstrap shows workload sensitivity, especially for two-recording MSL. Aggregate intervals adjust over three collections; individual semantics intervals adjust over 36 collection/semantics cells per comparison. The extreme adjusted quantiles of 10,000 draws are approximate. No host pooling or favorable-host selection is used. The original and supplemental native consumer intervals resample six paired complete-workflow trials only when all six are observed. The separate five-process follow-up instead resamples process-level mean log ratios, as specified in Table 8.

*Correctness and selection.* Complete canonical range curves are archived, including precision, recall and F1; repeated-output hashes must agree. Independent record auditors import no metric producer and reconstruct coverage, curve discrepancies, winner sets, regret and intervals. Finite-set rational oracles additionally cover exhaustive small tied inputs, with randomized boundary and mixed-weight cases. Public implementations and a long-window high-precision check provide further differential evidence. Real-arithmetic identities, empirical binary64 agreement and a formal rounding certificate are distinct.

Winner sets contain values within $10^{-10}$ of the maximum. Approximation regret is the best exhaustive value minus the selected

model's exhaustive optimum. The original panel retains minimum and maximum over tied winners; the weighted panel's prespecified primary statistic uses the best tied winner. Worst-tie sensitivity is a separate subsequent reconstruction. The predeclared practical cutoff is .01. This is labeled oracle evaluation, not a deployable threshold policy or an improvement in detector accuracy.

## 6 VUS AND ORIGINAL RANGE RESULTS

*VUS attribution.* At width 30, the whole stream evaluator improves over exact Klein by 25.17–30.51 times geometrically and over a matched compiled dense evaluator by 3.54–3.96 times (Table 1). The public-library advantage therefore includes implementation effects but is not entirely explained by them. Independent-call dense/released ratios are 2.37/2.91/3.28. All primary exact PR calls agree within $1.11 \cdot 10^{-15}$; matched ROC workflow differences are at most $4.86 \cdot 10^{-14}$. Against unchanged Benchopt with equal ROC+PR outputs, directly timed workflow ratios are 25.3/32.1/40.9.

The new sharing increment is smaller: released/stream ratios are 1.496/1.271/1.207. Geometry reuse already explains most of it on SMAP/SMD; the additional stream/geometry improvement is 1.094/1.024/1.025, with five SMAP/SMD case losses. The conventional compiled support control is faster than the released single-model evaluator on every primary case (ratios .910/.960/.978, adjusted intervals below one). Thus this panel does not establish a separate runtime advantage of nearest geometry over optimized direct ramps. Control/stream ratios remain 1.357/1.201/1.166, including shared work. The minimal Klein patch alone improves public Klein by 1.65/2.27/2.97; the remaining released/patch advantage combines additional reductions and implementation differences. These ablations preclude assigning the entire public-library gap to one geometric lemma.

At 250 sampled Klein thresholds, 14/73 winner sets change and four have regret at least .01 (maximum .0584); at 1,000, three change and none has regret .01. NAB has no changes. Complete stream workflows are nevertheless 7.27–11.25 times faster than the sampled 250-query public API. This comparison has different outputs and establishes an observed cost/fidelity tradeoff, not same-query dominance over every approximate algorithm.

*Failure regimes and broader VUS checks.* Primary support fractions are .00384–.223. In a controlled dense-positive case, dense/released is .980 with interval [.962,.9994]: sparse compression loses. A one-million-point random-score case visits 99.94% of support in the active-prefix algorithm, explaining why saturation is not a general solution. With one profile, sharing gains only 1.023 times. A six-series development certificate-plus-fallback route took 33.93 times the direct exact cost and certified no early selections; this failed route was not promoted.

Eight MITDB and one SWaT recording extend real lengths to 419,919–650,000. Complete Klein/stream workflow median ratios are 23.10 and 17.70; incremental released/stream paired intervals [.925,1.340] and [.955,1.165] include one. These reused follow-up recordings support the whole evaluator more strongly than the batch refinement. At $n = 500{,}000$, eight profiles and positive fraction .5, process peak RSS is 225.0 MiB for stream, versus released 175.2 and dense 210.6 MiB; lifetime process peaks include libraries, inputs and warm-up.

The earlier unchanged TSB-AD nine-metric consumer shows an Amdahl limit. On the length-selected SMD case with eight profiles, replacing only VUS reduces its stage from .9289 to .01406 seconds but total read-to-CSV time from 34.260 to 33.234 seconds: 1.031×. All nine outputs agree within $3.45 \cdot 10^{-15}$. Two complete repetitions per backend give descriptive timing, not a population interval. Affiliation and other stages dominate. The new three-metric TimeEval experiment is a separately defined consumer, not a revision of this unfavorable observation.

*Original flat/reciprocal range trajectory.* Table 2 compares exhaustive observed-threshold evaluation against a fused compiled raster, including preparation and winner selection. The maximum curve difference is $3.00 \cdot 10^{-15}$, with no method-induced winner disagreement. Bucketing wins every grid100 case median; the exhaustive sweep loses on SMAP534, whose eight profiles have only 3–11 distinct values. Its original Apple raster/sweep ratio is .417, and its fresh Linux ratio .661. Repeated scanning is competitive when there are very few queries.

Replacing grid100 by complete strict observed thresholds changes 24/73 winner sets, with 15 disjoint sets and 12 best-selected-model regrets at least .01. Maximum Apple regrets are .439/.150/.127. These are consequences within a reused panel of eight authored score procedures. They do not establish a general prevalence of harmful approximation. Complete sweep costs 2.19/2.14/3.01 times the fastest grid100 bucket method on the original host; the extra fidelity is not free.

Length-selected public comparisons use the three shortest cases per collection, all eight profiles, and a declared 60-second per-call budget. TSB-AD grid-F1 values agree within $6.67 \cdot 10^{-16}$; completed TimeSeAD calls agree within $5.66 \cdot 10^{-8}$ under its float32 storage and separate $\alpha = 0$ contract. Two public SMD187 calls are censored and are not counted as parity successes. All 1,168 unchanged-public binary range-F1 checks at selected cuts agree within $4.62 \cdot 10^{-16}$; this validates chosen-cut values, not independent global maximality. Three million-point stress cases have maximum cross-method curve error $2.23 \cdot 10^{-12}$, below the frozen $10^{-9}$ stress tolerance.

*Fresh original-panel execution.* A complete raw-data-to-table replay on an AMD EPYC 7763 independently regenerates profiles and reproduces the direction of the VUS and range results. Original VUS released/stream ratios are 1.886/1.580/1.340 and Klein workflow ratios 20.06/20.28/16.17. The combined range replay retains all generated arrays and curves, and its four independent reports reconstruct identically. All 73 score matrices differ bitwise from the Mac; differences are not uniformly small. The range selection counts remain 24/15/12, but two SMD recordings exchange flags and maximum SMD regret becomes .0797. A separate SVM diagnostic finds 104 versus 105 fitted support vectors despite identical raw and standardized inputs; it does not identify a specific instruction-level cause. Fresh generation is therefore not an isolated CPU intervention.

| Collection | Cases | Released / stream | Matched dense / stream | Klein exact / stream |
|---|---|---|---|---|
| NAB | 20 | 1.496 [1.428,1.566] | 3.543 [3.338,3.713] | 30.513 [27.399,33.652] |
| SMAP | 17 | 1.271 [1.242,1.302] | 3.693 [3.196,4.191] | 25.772 [22.941,28.940] |
| SMD | 36 | 1.207 [1.193,1.221] | 3.956 [3.736,4.125] | 25.172 [23.475,26.695] |

**Table 1: Original VUS complete-call geometric ratios, Apple M1 Pro, width 30, eight profiles per case. Brackets are 95% paired-case bootstrap bounds adjusted over three collections per comparator. They characterize the fixed panel; source and label dependence remain. All calls include validation, preparation and winner selection. This is the whole evaluator versus its released precursor, a matched compiled dense control, and the unchanged public exact API.**

| Host / collection | Grid100 raster/bucket | Distinct raster/sweep | Distinct wins/cases |
|---|---|---|---|
| Apple NAB | 1.719 [1.625,1.814] | 27.80 [18.74,41.94] | 20/20 |
| Apple SMAP | 1.994 [1.817,2.252] | 21.44 [8.57,43.14] | 16/17 |
| Apple SMD | 1.969 [1.884,2.095] | 128.08 [95.09,154.61] | 36/36 |
| Linux NAB | 1.900 [1.616,2.239] | 29.35 [18.81,46.08] | 20/20 |
| Linux SMAP | 2.898 [2.624,3.260] | 27.76 [11.63,53.75] | 16/17 |
| Linux SMD | 3.574 [3.412,3.789] | 163.60 [125.92,193.02] | 36/36 |

**Table 2: Original flat/reciprocal range-F1 study on the same 73 recordings and 584 profiles. Ratios compare identical query contracts against a fused compiled raster. Familywise 95% intervals adjust over three collections separately per comparison. Linux is the selected fresh combined EPYC 7763 execution; generated scores differ and hosts are not pooled. The exhaustive low-cardinality loss is retained on both hosts.**

## 7 WEIGHTED AND NATIVE CONSUMER RESULTS

*Generalized operator on separately selected recordings.* The completed Mac study contains all 26 recordings, both public detector profiles, 12 bias/cardinality combinations, both query contracts and all scheduled methods. The independent reconstruction finds maximum curve discrepancy $2.50 \cdot 10^{-15}$ and no same-contract winner disagreement. Table 3 reports equal-semantic-weight geometric ratios against the fused compiled controls. Grid bucketing improves every recording's aggregate on all three collections. Exhaustive sweep improves all WSD and Stock aggregates but only one of two MSL recordings. MSL's exhaustive ratios are .558 and 3.438, so its narrow paired-timing interval around the aggregate does not indicate robust benefit across new MSL recordings. The separate recording-reweighting interval [.558,3.438] exposes this sensitivity. The 12 individual semantic contracts retain the same collection-level direction: their exhaustive ratios span 8.388–8.959 (WSD), 20.988–23.321 (Stock), and 1.313–1.449 (MSL). Their grid ratios span 1.506–1.549, 1.472–1.714, and 1.519–1.592, respectively. All adjusted paired-order intervals for these aggregate semantic cells lie above one. Nevertheless, every MSL exhaustive contract loses on the same one of its two recordings; the semantic average does not erase that loss. The artifact retains all 36 collection/semantics rows per comparator with both uncertainty summaries.

Sampling has a smaller selection effect on this panel. Only the middle-bias, cardinality-one contract produces a nonoverlapping grid-versus-exhaustive winner set, with maximum best-tie regret .001412; no contract produces best-tie regret .01. A separate post-hoc saved-curve reconstruction finds singleton grid winners in all 312 case/semantic combinations, so worst-tie regret is identical on this Mac panel. Thus the original 15/73 nonoverlap result does not extend as a prevalence claim to these two unchanged public detectors. The weighted study supports computational generality while retaining this negative selection result.

*Fresh Linux execution.* The original fresh replay verifies the raw archive, 425 TimeEval source files and all typed inputs, regenerates 52 detector profiles, and reconstructs both operator and consumer outputs. Compared with Mac, all 26 label-array hashes agree but 25/52 score hashes differ, all from COPOD; FFT's 26 score hashes agree. These are independently generated host workloads, not an isolated CPU intervention. Native query totals are 2,776 at the nominal default on both hosts and 71,749/71,750 at all observed queries on Mac/Linux. No timing or uncertainty is pooled across hosts. The Linux weighted study reconstructs all 11,232 records and 156 curves, with maximum error $2.50 \cdot 10^{-15}$ and zero same-contract winner disagreements. Table 3 gives its aggregate results. Linux MSL id7 loses against exhaustive raster (.859) and against grid sweep (.963); ten of the MSL collection-level grid sweep/bucket semantic intervals include one. Thus the Mac semantic result is not asserted uniformly across Linux contracts. All 312 Linux grid winner sets are also singletons; the largest best/worst-tie regret remains .001412, with none reaching .01.

*Native public consumer.* The Mac execution completes 120 of 144 scheduled records, with four public 600-second censors and 20 prespecified skips. The independent auditor verifies all completed CSVs and reconstructs maximum scalar discrepancy $3.33 \cdot 10^{-16}$. Completed public outputs cover 28/52 profiles in each contract (MSL and WSD); Stock has compiled-to-compiled agreement only. Matching family outputs also verify those same profiles in the combined batch, but do not supply its censored public execution time.

| Host | Collection | Grid100 raster/bucket | Distinct raster/sweep | Distinct recording sensitivity |
|---|---|---|---|---|
| Apple | WSD | 1.526 [1.518,1.530] | 8.801 [8.727,8.837] | [4.896,14.920] |
| Apple | Stock | 1.621 [1.609,1.628] | 22.635 [22.424,22.810] | [17.181,29.800] |
| Apple | MSL | 1.559 [1.542,1.573] | 1.385 [1.362,1.427] | [.558,3.438] |
| Linux | WSD | 2.097 [2.081,2.101] | 7.460 [7.367,7.484] | [4.290,12.454] |
| Linux | Stock | 1.738 [1.734,1.742] | 15.848 [15.628,15.875] | [13.043,19.373] |
| Linux | MSL | 2.072 [2.045,2.090] | 1.670 [1.659,1.682] | [.859,3.244] |

**Table 3: Original weighted operator study, Apple M1 Pro and fresh Linux replay. Within each recording, log ratios are equally weighted across the 12 prespecified bias/cardinality contracts. Bracketed ratio columns give approximate 95% paired-order intervals conditional on this fixed panel, adjusted over three collections. The last column separately resamples recordings and is workload sensitivity, not a population confidence interval. WSD and Stock each have 12 recordings; MSL has two. All ratios compare identical outputs and queries within a row.**

Table 4 shows absolute whole-workflow medians and matched compiled comparisons. At all observed queries, the combined raster/bucket ratio is 2.728 [2.649,2.801], including native loading, three metrics and result writing. At the nominal default it is 1.017 [.998,1.039], with no supported advantage over raster. Public-code engineering gains are much larger: completed WSD public/bucket ratios are 9.598 [9.438,9.774] at the default and 407.099 [398.649,416.713] at all observed queries. These are not all attributable to the incremental recurrence. The separately frozen interval control below tests the remaining conventional-algorithm explanation.

The original fresh Linux replay also completes 120 workflows, with four 600-second public censors and 20 prescribed skips. It verifies public outputs for the same 28/52 profiles per contract, with maximum discrepancy $3.33 \cdot 10^{-16}$. Table 5 retains all batch results. The combined exhaustive raster/bucket ratio is 1.932 [1.907,1.946], smaller than the separately executed conventional-control replay below. Its default ratio is 1.014 [1.007,1.027], a small benefit that is not consistently supported by the other runs' intervals. MSL's exhaustive aggregate does not support a gain; one bucket trial takes .09497 seconds, versus .04342–.04491 for its other five trials. That observation is retained, not removed as an outlier.

*Conventional interval control.* The separately frozen ablation compares incremental curves with closed-form interval recomputation and remeasures raster on the same panel. Its fresh Linux run regenerates all 52 public profiles before timing. Both hosts complete all scheduled weighted cases and 144 native workflows. Maximum weighted curve discrepancy is $2.39 \cdot 10^{-15}$ on each host; native scalar discrepancy is at most $3.33 \cdot 10^{-16}$ on each host. Table 6 gives the interval comparisons. Every collection/query aggregate favors incremental evaluation, but interval is not uniformly the fastest conventional control: raster/interval aggregate ratios are below one in five of six Mac comparisons and all six Linux comparisons. Both controls remain visible.

On Mac, all 72 collection/semantic/query intervals against interval control lie above one, but MSL id7 loses every exhaustive semantic setting (aggregate .775), while id3 improves 3.759 times. On Linux, id7 improves only 1.024 times over interval; the faster raster still limits the benefit. The exhaustive interval/sweep recording-sensitivity intervals are [4.290,11.919], [19.472,35.575] and [.775,3.759] on Mac, and [7.566,21.139], [27.994,46.863] and [1.024,5.536] on Linux, in WSD/Stock/MSL order. The conditional aggregate intervals therefore do not establish uniformly positive performance on new recordings.

A separate exploratory post-hoc contrast directly reconstructs raster over incremental from the same supplemental trials, with paired-order tails adjusted over all six collection/query contrasts per host (Table 6). These exploratory intervals do not correct post-hoc comparator selection. The fixed-raster contrast retains MSL id7's loss on both hosts; Figure 1 also displays the faster of the two conventional controls per recording without assigning an interval to this lower envelope.

The native conventional-control ablation retains complete three-metric read-to-write semantics with flat weights and cardinality one for range PR-AUC; the broader weighted family is evaluated by the operator study. At all observed queries, combined interval/bucket ratios are 3.034 [2.990,3.075] on Mac and 3.190 [3.166,3.224] on Linux. The faster combined raster control gives 2.732 [2.656,2.812] and 2.756 [2.742,2.773], respectively. At default sampling, raster/bucket is only 1.026 [.998,1.049] and 1.016 [.997,1.033]. Thus the exhaustive-workflow gain survives the additional control on both hosts, while neither host supports a default combined-workflow gain over raster. These compiled-only trials add no public-reference coverage to the earlier study.

Across all four original/supplemental Apple/Linux executions, the combined exhaustive raster/bucket ratio spans 1.932–2.756; results are not pooled. The two Linux runs have identical hashes for all 52 score profiles and the shared sources, but their recorded OpenBLAS dispatch architectures differ (Cooperlake versus SkylakeX). The supplemental combined raster median is 35.2% higher and bucket median 4.3% lower than in the original run. These descriptive differences do not identify a causal CPU or load effect. The narrow paired-order intervals are conditional within each run and do not cover this between-run variability.

*Five process-session follow-up.* A strict prior-score replication attempt stopped before any metric or warm-up: 25 COPOD profiles differed in score hashes and one changed its distinct-score count. We preserved that failed v1 and prospectively froze a separate v2 using the same recordings and exactly one new 52-profile generation attempt, with strict raw/label/template identity and prior score/query differences retained as diagnostics. V2's first generation happened to match all prior Linux score and query-plan

| Query contract | Batch (profiles) | Public seconds | Raster seconds | Bucket seconds | Raster/bucket [95% interval] |
|---|---|---|---|---|---|
| Nominal 50 | All (52) | censored > 600 | 1.0848 | 1.0783 | 1.017 [.998,1.039] |
| Nominal 50 | MSL (4) | .08008 | .03177 | .03291 | .860 [.639,1.014] |
| Nominal 50 | Stock (24) | censored > 600 | .48931 | .48169 | 1.043 [1.013,1.083] |
| Nominal 50 | WSD (24) | 5.22731 | .55475 | .54321 | 1.019 [1.003,1.037] |
| All observed | All (52) | censored > 600 | 2.94605 | 1.08068 | 2.728 [2.649,2.801] |
| All observed | MSL (4) | .48152 | .03352 | .03038 | 1.088 [1.045,1.136] |
| All observed | Stock (24) | censored > 600 | 1.86034 | .50496 | 3.557 [3.354,3.715] |
| All observed | WSD (24) | 234.80063 | 1.07523 | .58478 | 1.866 [1.832,1.901] |

**Table 4: Native TimeEval read-to-result-write execution on Apple M1 Pro. Absolute times are medians of six completed workflows. Ratios are geometric means of six matched-order ratios, not ratios of the displayed medians; intervals are approximate conditional paired-order intervals. Each censored public key has one timeout and five unrun repetitions, not a completed median or interval. The native nominal/all-observed contracts emit 2,776/71,749 queries across 52 profiles; no cross-contract speed ratio is asserted.**

| Query contract | Batch (profiles) | Public seconds | Raster seconds | Bucket seconds | Raster/bucket [95% interval] |
|---|---|---|---|---|---|
| Nominal 50 | All (52) | censored > 600 | 1.58408 | 1.56792 | 1.014 [1.007,1.027] |
| Nominal 50 | MSL (4) | .25969 | .04442 | .04485 | .992 [.986,.9997] |
| Nominal 50 | Stock (24) | censored > 600 | .71023 | .70782 | 1.020 [1.003,1.046] |
| Nominal 50 | WSD (24) | 14.44195 | .83543 | .82163 | 1.027 [1.015,1.050] |
| All observed | All (52) | censored > 600 | 3.04217 | 1.56594 | 1.932 [1.907,1.946] |
| All observed | MSL (4) | 1.94251 | .04737 | .04445 | .941 [.730,1.076] |
| All observed | Stock (24) | censored > 600 | 1.68294 | .70907 | 2.342 [2.280,2.377] |
| All observed | WSD (24) | 406.24099 | 1.33226 | .84578 | 1.577 [1.559,1.593] |

**Table 5: Original public-reference TimeEval replay on Linux. Timing, ratio, censor and uncertainty definitions match Table 4; native query totals are 2,776/71,750. Stock and combined public calls have no completed median. All completed trials, including the slower MSL bucket trial, enter the reported estimates.**

| | | Grid100: control/bucket | | Distinct: control/sweep | |
|---|---|---|---|---|---|
| Host | Collection | Raster† | Interval | Raster† | Interval |
| Apple | WSD | 1.526 [1.518,1.531] | 1.665 [1.658,1.671] | 8.608 [8.549,8.639] | 7.323 [7.277,7.342] |
| Apple | Stock | 1.611 [1.602,1.618] | 2.051 [2.040,2.059] | 21.991 [21.908,22.050] | 26.339 [26.244,26.387] |
| Apple | MSL | 1.571 [1.552,1.584] | 1.753 [1.725,1.759] | 1.421 [1.403,1.461] | 1.706 [1.679,1.728] |
| Linux | WSD | 1.672 [1.661,1.680] | 2.308 [2.293,2.315] | 11.621 [11.111,11.668] | 12.945 [12.396,12.991] |
| Linux | Stock | 1.548 [1.543,1.555] | 2.289 [2.282,2.296] | 26.717 [26.579,26.786] | 36.149 [35.977,36.224] |
| Linux | MSL | 1.665 [1.654,1.690] | 2.331 [2.306,2.348] | 1.726 [1.695,1.776] | 2.381 [2.347,2.418] |

**Table 6: Both conventional controls in the supplemental weighted study. All methods share compiler and output semantics. Interval uses closed-form overlap at each query, with no cross-query metric state. Brackets quantify conditional paired-order timing uncertainty, not workload generalization. Interval contrasts were prospectively frozen and adjust over three collections per query contract. †Raster contrasts are exploratory post-hoc reconstructions adjusted over six collection/query contrasts per host; this does not correct post-hoc comparator selection. No hosts are pooled. All semantic results and recording-reweighting intervals remain in the artifact.**

hashes; matching was not a selection criterion. Five new processes shared that frozen realization and one permitted logical CPU, with fixed full-workflow warm-ups and balanced method orders. Table 8 retains every session estimate. All 180 timed workflows complete; saved scalar discrepancy across methods, warm-ups and sessions is at most $3.33 \cdot 10^{-16}$. The exhaustive raster/bucket ratio is 1.919, with session estimates 1.905–1.933; interval/bucket is 2.216. The nominal-default raster gain is only 1.017. Native query totals remain 2,776/71,750. This supports fixed-input process repeatability; it does not explain the earlier run difference, establish new-host generalization, or add public-reference coverage. No earlier result is replaced or pooled.

Table 7 gives a descriptive cost–fidelity observation, rather than a same-output speed ratio. Moving from the native nominal default to all observed queries increases query count from 2,776 to 71,749/71,750 on Mac/Linux, while the incremental combined workflow's median changes from 1.043 to 1.073 seconds and from

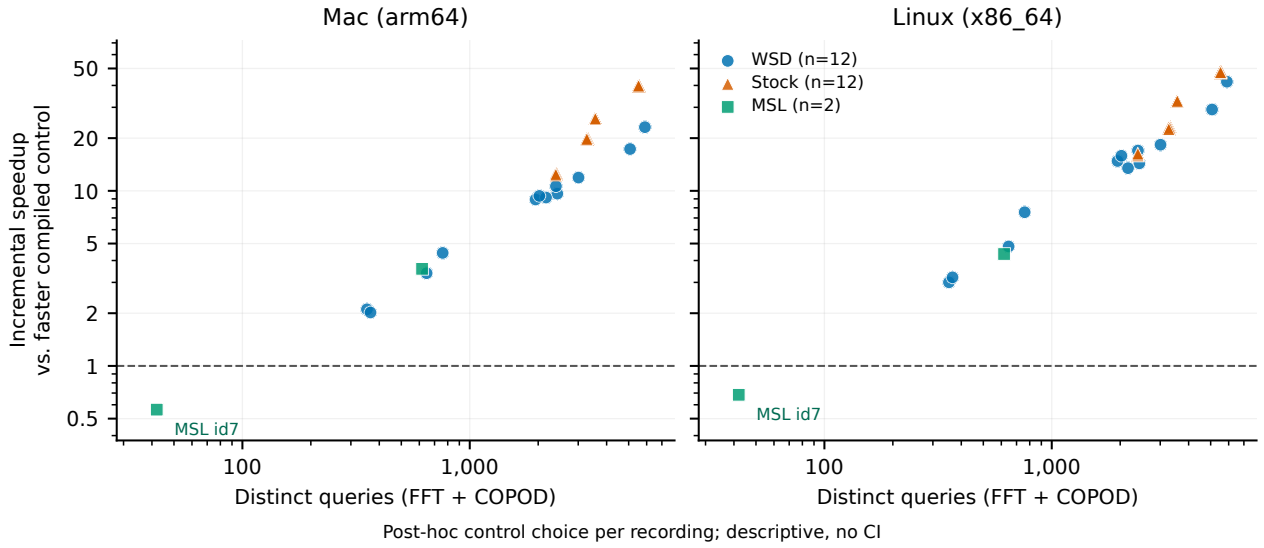



**Figure 1: Post-hoc descriptive query-regime view of the supplemental study. Each point is one recording, equally weighting its 12 semantic log ratios. The denominator is incremental evaluation; the numerator selects the faster geometric aggregate of raster and interval for that recording. This is a descriptive lower envelope of measured controls, not a deployable selector or a prespecified confidence interval. MSL id7 loses on both hosts. Distinct query counts sum the two public detector profiles.**

| Execution | Queries | Raster s | Interval s | Bucket s | Raster/bucket | Interval/bucket |
|---|---|---|---|---|---|---|
| Apple O | Nominal 50 | 1.0848 | — | 1.0783 | 1.017 [.998,1.039] | — |
| Apple O | All observed | 2.9460 | — | 1.0807 | 2.728 [2.649,2.801] | — |
| Linux O | Nominal 50 | 1.5841 | — | 1.5679 | 1.014 [1.007,1.027] | — |
| Linux O | All observed | 3.0422 | — | 1.5659 | 1.932 [1.907,1.946] | — |
| Apple C | Nominal 50 | 1.0686 | 1.0991 | 1.0429 | 1.026 [.998,1.049] | 1.043 [1.016,1.067] |
| Apple C | All observed | 2.9474 | 3.2754 | 1.0726 | 2.732 [2.656,2.812] | 3.034 [2.990,3.075] |
| Linux C | Nominal 50 | 1.4617 | 1.4880 | 1.4381 | 1.016 [.997,1.033] | 1.034 [1.015,1.047] |
| Linux C | All observed | 4.1126 | 4.7551 | 1.4990 | 2.756 [2.742,2.773] | 3.190 [3.166,3.224] |

**Table 7: All four combined native TimeEval executions: O denotes the original public-reference study and C the supplemental conventional-control study. Each row evaluates all 52 profiles and all three metrics from CSV read to result write/close; range PR-AUC uses flat weights and cardinality one. Absolute times are medians of six runs; paired ratios and conditional intervals are reconstructed from matched orders within each execution. No runs are pooled. C adds no public-reference coverage.**

| Query contract | Control / bucket | Session 1 | 2 | 3 | 4 | 5 | Geometric ratio [descriptive interval] |
|---|---|---|---|---|---|---|---|
| Nominal 50 | Raster | 1.010 | 1.013 | 1.017 | 1.016 | 1.028 | 1.017 [1.011,1.025] |
| Nominal 50 | Interval | 1.019 | 1.018 | 1.017 | 1.025 | 1.034 | 1.023 [1.017,1.030] |
| All observed | Raster | 1.907 | 1.905 | 1.931 | 1.933 | 1.919 | 1.919 [1.906,1.932] |
| All observed | Interval | 2.195 | 2.209 | 2.226 | 2.229 | 2.221 | 2.216 [2.201,2.227] |

**Table 8: Prospectively frozen v2 follow-up: five fresh processes on one AMD EPYC 9V74 runner, pinned to the same permitted logical CPU. Each session estimate averages six paired log ratios; the final estimate equally weights the five sessions. Intervals enumerate all $5^5$ empirical resamples with tails adjusted over four contrasts. They describe this fixed-input, shared-runner experiment, without guaranteed coverage or new-host inference. All 180 timed workflows and 30 full untimed warm-ups are retained.**

1.438 to 1.499 seconds. This workload can therefore obtain the complete native query close to its sampled workflow cost. These are warm-workflow medians under different query contracts, not paired timing intervals or evidence of a practically important detector-selection improvement.

The per-collection native results retain practical counterexamples. At default sampling, Linux MSL favors raster (.981 [.973,.988] raster/bucket), while Stock's .997 [.967,1.018] interval crosses one. At all observed queries, interval/bucket on Linux is 1.256 [1.090,1.643] for MSL, 4.649 [4.632,4.666] for Stock and 1.897 [1.849,1.932] for WSD. The conventional-control ordering for exhaustive WSD changes across hosts: raster/interval is 1.073 on Mac but .961 on Linux. Reporting only the slower conventional implementation would overstate the joint-state increment.

## 8 RELATED WORK AND NOVELTY BOUNDARY

The original VUS paper and its effective/efficient extension are the closest database metric antecedents [2, 18]. Our formulas preserve their dynamic target. Sylligardos [22], Klein and Benchopt establish that fast exact evaluation and support cumsums are not new objectives. The source audit distinguishes current definitions before forming speed ratios. Tatbul [24] already defines positional weights, cardinality penalties and closed-form per-overlap evaluation. TimeSeAD already enumerates observed thresholds, and TimeEval already offers weighted PR curves. Classical component trees maintain connected level sets and attributes; linear-time one-dimensional construction predates this work [14, 16]. Naegel and Passat [15] also incrementally compute component attributes relative to a fixed binary segmentation target. Label conditioning itself is therefore established. Ranefall and Wählby already derive per-threshold object precision/recall from component-size attributes [20]. Their true-positive pixels are defined by a size interval and their recall denominator by the maximum such count, rather than fixed labeled anomaly intervals. These constructions supply related threshold/attribute frameworks. Our additional state must handle label intersections, fragmentation, and moving weight normalization. The additional identity maintains *joint label-conditioned overlap and fragmentation aggregates*, including moving precision weights, across a trajectory. It is not a new component tree or a new weighting scheme.

The 2024–2026 comparison also includes distinct nearby research questions. PATE [6] introduces proximity-aware evaluation, while we optimize existing targets. TSB-AD [12] and TAB [19] address benchmark reliability and unification; ESA [10] contributes telemetry benchmark conditions. These motivate reusable evaluation operators rather than establish our state representation. TSB-AutoAD [11] and RAMSAD [17] study selection, ensembling or retrieval of detector profiles; we make no superior unlabeled selection claim. mTSBench [32] covers a broader multivariate workload, and the one-liner foundation-model analysis [33] studies detection pipelines rather than incremental metric state.

Formal analysis and LARM/ALARM [29] examine metric axioms and exclude score-based VUS from their formal scope. Their properties do not prove this computational identity. CCE offers a different linear-time metric [31]; its native runtime is not an equal-output baseline. A problem-oriented taxonomy [30] and adversarial reporting analysis [13] address suitability and reporting choices. Rank changes themselves are not a new discovery. Our bounded literature/source search supports the narrower local-state contribution, not universal precedence over every unpublished implementation.

## 9 REPRODUCIBILITY AND LIMITS

Raw archives and members, decoded arrays, generated scores, source revisions, protocols and outputs are SHA-256 bound. The primary interval-control and compiled-consumer results are produced from fresh raw data and public sources with:

```
python experiments/reproduce_interval_control.py
  --output /tmp/interval-control-replica
```

The destination must be new; dependencies and observed versions are supplied. This command fits public profiles, verifies TimeEval loaders, executes both studies and reconstructs primary tables. The guide supplies separate saved-record commands for exploratory raster intervals. Original weighted/public-consumer, five-process and earlier VUS/range reproductions respectively use `reproduce_strengthening.py`, `reproduce_affinity_blocks_v2.py --run`, and `reproduce_range_sweep.py --include-vus`, each with a new destination. Supplementary studies retain separate protocols; TeX compilation is distinct from numerical reproduction. New raw datasets and score vectors remain local because redistribution terms are unconfirmed. Acquisition instructions, identifiers, digests, metric curves and result CSVs are public.

Mathematical claims assume unit-cost exact metric arithmetic. Binary64 experiments give empirical parity, not an all-input rounding certificate. Canonical integer moments have an explicit size bound; arbitrary polynomial weights are theorem-only. Native TimeEval integration covers its declared flat/cardinality-one contract, not every callback option. Low query counts can favor raster scans, dense VUS support can remove compression gains, and shared-label batching can increase memory. Warm stored-profile workloads do not measure detector training, cold compilation or storage durability. Sources and regenerated scores differ across recorded environments, and repeated recordings or timing orders are not independent domains. These limits define where the exact operator acceleration is supported.

*Tool assistance.* AI-assisted tools, including OpenAI Codex, were used for literature search, mathematical and software development, experiment orchestration, independent implementation checks, and drafting. The artifact records the validation procedures and source/result provenance. The sole human author is responsible for the work; automated audits and simulated reviews are not external peer review.

## 10 CONCLUSION

Dynamic VUS and weighted range metrics admit different local sufficient statistics. Complete ties and predecessor rectangles preserve VUS integrals; coupled overlap and fragmentation updates preserve weighted range trajectories. Compilation-matched controls, separate public-detector workloads and native consumers establish the exact operators' measured operating range and expose their low-query limits.

## REFERENCES

[1] Benchopt contributors. 2026. Time-series foundation-model benchmark: VUS metrics. https://github.com/benchopt/benchmark_tsfm Software, audited

revision 2b55ccb1, 29 September 2026.

[2] Paul Boniol, Ashwin K. Krishna, Marine Bruel, Qinghua Liu, Mingyi Huang, Themis Palpanas, Ruey S. Tsay, Aaron Elmore, Michael J. Franklin, and John Paparrizos. 2025. VUS: Effective and Efficient Accuracy Measures for Time-Series Anomaly Detection. *The VLDB Journal* 34, Article 32 (2025), 26 pages. https://doi.org/10.1007/s00778-025-00907-x

[3] TimeSeAD contributors. 2025. TimeSeAD classic range F-score evaluator. https://github.com/wagner-d/TimeSeAD/blob/bb8efa9ce17f9cfb0bab77df08c32d986de8429e/timesead/evaluation/evaluator.py Source revision bb8efa9ce17f9cfb0bab77df08c32d986de8429e, inspected 24 October 2026.

[4] Erik Demaine. 2011. Lecture 7: Linear-Time Sorting. MIT OpenCourseWare, 6.006 Introduction to Algorithms, Fall 2011. https://ocw.mit.edu/courses/6-006-introduction-to-algorithms-fall-2011/bf7d79105762bf79bbc0925438e1468a_MIT6_006F11_lec07.pdf Accessed 2026-10-21.

[5] Tom Fawcett. 2006. An Introduction to ROC Analysis. *Pattern Recognition Letters* 27, 8 (2006), 861–874. https://doi.org/10.1016/j.patrec.2005.10.010

[6] Ramin Ghorbani, Marcel J. T. Reinders, and David M. J. Tax. 2024. PATE: Proximity-Aware Time series anomaly Evaluation. arXiv:2405.12096 [cs.LG] https://arxiv.org/abs/2405.12096

[7] Michael Herf. 2001. Radix Tricks. https://stereopsis.com/radix.html Accessed 2026-10-21.

[8] Tom Louis Klein. 2026. fast-VUS. https://github.com/tomlouisklein/fast-VUS Software, first commit 12 October 2026; audited revision 1bae859f, 16 October 2026.

[9] Youngmin Ko. 2026. Sparse VUS, version 0.3.0. https://github.com/mandu5/sparse-vus/releases/tag/v0.3.0 Public software and working-manuscript precursor, released 21 October 2026; source revision 68d6e6de627a734880e92ecc94d3ffc9eb38a89d.

[10] Krzysztof Kotowski, Christoph Haskamp, Jacek Andrzejewski, Bogdan Ruszczak, Jakub Nalepa, Daniel Lakey, Peter Collins, Aybike Kolmas, Mauro Bartesaghi, Jose Martinez-Heras, and Gabriele De Canio. 2024. European Space Agency Benchmark for Anomaly Detection in Satellite Telemetry. https://arxiv.org/abs/2406.17826 arXiv:2406.17826, revised 17 August 2025.

[11] Qinghua Liu, Seunghak Lee, and John Paparrizos. 2025. TSB-AutoAD: Towards Automated Solutions for Time-Series Anomaly Detection. *Proceedings of the VLDB Endowment* 18, 11 (2025), 4364–4379. https://doi.org/10.14778/3749646.3749699

[12] Qinghua Liu and John Paparrizos. 2024. The Elephant in the Room: Towards a Reliable Time-Series Anomaly Detection Benchmark. In *Advances in Neural Information Processing Systems*, Vol. 37. 108231–108261. https://github.com/TheDatumOrg/TSB-AD

[13] Zongye Lyu. 2026. Did We Actually Fix It? An Independent Adversarial Stress-Test of Post-Point-Adjustment Evaluation Metrics for Time-Series Anomaly Detection. arXiv:2607.11969 [stat.ML] https://arxiv.org/abs/2607.11969

[14] David Menotti, Laurent Najman, and Arnaldo de Albuquerque Araújo. 2007. 1D Component Tree in Linear Time and Space and Its Application to Gray-Level Image Multithresholding. In *Proceedings of the 8th International Symposium on Mathematical Morphology*. INPE, 437–448. https://hal.science/hal-00622373

[15] Benoît Naegel and Nicolas Passat. 2014. Interactive Segmentation Based on Component-trees. *Image Processing On Line* 4 (2014), 89–97. https://doi.org/10.5201/ipol.2014.71

[16] Laurent Najman and Michel Couprie. 2006. Building the Component Tree in Quasi-Linear Time. *IEEE Transactions on Image Processing* 15, 11 (2006), 3531–3539. https://doi.org/10.1109/TIP.2006.877518

[17] Christos Panourgias, Roberto Stanzione, Adrien Petralia, Themis Palpanas, and Paul Boniol. 2026. Retrieval Over Training: Similarity search-based Model Selection for Time Series Anomaly Detection. arXiv:2610.20243 [cs.LG] https://arxiv.org/abs/2610.20243

[18] John Paparrizos, Paul Boniol, Themis Palpanas, Ruey S. Tsay, Aaron Elmore, and Michael J. Franklin. 2022. Volume Under the Surface: A New Accuracy Evaluation Measure for Time-Series Anomaly Detection. *Proceedings of the VLDB Endowment* 15, 11 (2022), 2774–2787. https://doi.org/10.14778/3551793.3551830

[19] Xiangfei Qiu, Zhe Li, Wanghui Qiu, Shiyan Hu, Lekui Zhou, Xingjian Wu, Zhengyu Li, Chenjuan Guo, Aoying Zhou, Zhenli Sheng, Jilin Hu, Christian S. Jensen, and Bin Yang. 2025. TAB: Unified Benchmarking of Time Series Anomaly Detection Methods. arXiv:2506.18046 [cs.LG] https://arxiv.org/abs/2506.18046

[20] Petter Ranefall and Carolina Wählby. 2016. Global gray-level thresholding based on object size. *Cytometry Part A* 89, 4 (2016), 385–390. https://doi.org/10.1002/cyto.a.22806

[21] scikit-learn developers. 2021. Precision–Recall Curve, version 1.0.2. https://github.com/scikit-learn/scikit-learn/blob/1.0.2/sklearn/metrics/_ranking.py Full-recall truncation in precision_recall_curve; inspected 2026-10-21.

[22] Emmanouil Sylligardos. 2025. Fast or Accurate? Rethinking Time Series Anomaly Detection. In *VLDB Ph.D. Workshop*. https://www.vldb.org/2025/Workshops/VLDB-Workshops-2025/PhD/PhD25_17.pdf Section 4.

[23] Emmanouil Sylligardos. 2026. FF-VUS implementation in aVUSi. https://github.com/doanthihoaithu/aVUSi Software, audited revision facf4362, 23 June 2026.

[24] Nesime Tatbul, Tae Jun Lee, Stan Zdonik, Mejbah Alam, and Justin Gottschlich. 2018. Precision and Recall for Time Series. In *Advances in Neural Information Processing Systems*. https://papers.neurips.cc/paper_files/paper/2018/hash/8f468c873a32bb0619eaeb2050ba45d1-Abstract.html

[25] TheDatumOrg. 2026. VUS: Official Metric Implementation. https://github.com/TheDatumOrg/VUS/blob/467bace0c103f5c2bdb4900306d82939bcac8ff9/vus/utils/metrics.py Pinned revision 467bace0; RangeAUC_volume_opt with exhaustive thresholds, inspected 2026-10-21.

[26] TimeEval contributors. 2022. Optimize range-based metrics. https://github.com/TimeEval/TimeEval/issues/22 Issue 22; accessed 24 October 2026.

[27] TimeEval contributors. 2026. TimeEval: Range Precision–Recall Curve Implementation. https://github.com/TimeEval/TimeEval/blob/7c3622757b1a3afa9a84cbc97ce132d9e379f715/timeeval/metrics/range_metrics.py Software, audited revision 7c3622757b1a3afa9a84cbc97ce132d9e379f715; accessed 24 October 2026.

[28] Dennis Wagner, Tobias Michels, Florian C. F. Schulz, Arjun Nair, Maja Rudolph, and Marius Kloft. 2023. TimeSeAD: Benchmarking Deep Multivariate Time-Series Anomaly Detection. *Transactions on Machine Learning Research* (2023). https://openreview.net/forum?id=iMmsCI0JsS

[29] Dennis Wagner, Arjun Nair, Billy Joe Franks, Justus Arweiler, Aparna Muraleedharan, Indra Jungjohann, Fabian Hartung, Mayank C. Ahuja, Andriy Balinskyy, Saurabh Varshneya, Nabeel Hussain Syed, Mayank Nagda, Phillip Liznerski, Steffen Reithermann, Maja Rudolph, Sebastian Vollmer, Ralf Schulz, Torsten Katz, Stephan Mandt, Michael Bortz, Heike Leitte, Daniel Neider, Jakob Burger, Fabian Jirasek, Hans Hasse, Sophie Fellenz, and Marius Kloft. 2025. Formally Exploring Time-Series Anomaly Detection Evaluation Metrics. arXiv:2510.17562 [cs.LG] https://arxiv.org/abs/2510.17562

[30] Kaixiang Yang, Jiarong Liu, Yupeng Song, Shuanghua Yang, and Yujue Zhou. 2026. A Problem-Oriented Taxonomy of Evaluation Metrics for Time Series Anomaly Detection. arXiv:2511.18739 [cs.AI] https://arxiv.org/abs/2511.18739

[31] Zhijie Zhong, Zhiwen Yu, Yiu ming Cheung, and Kaixiang Yang. 2025. CCE: Confidence-Consistency Evaluation for Time Series Anomaly Detection. arXiv:2509.01098 [cs.LG] https://arxiv.org/abs/2509.01098

[32] Xiaona Zhou, Constantin Brif, and Ismini Lourentzou. 2026. mTSBench: Benchmarking Multivariate Time Series Anomaly Detection and Model Selection at Scale. *Transactions on Machine Learning Research* (2026). https://openreview.net/forum?id=8LfB8HD1WU Code revision 0ebf90da; dataset revision 8dea9af6.

[33] Xiaokun Zhu, Louis Carpentier, and Mathias Verbeke. 2026. When Foundation Models are One-Liners: Limitations and Future Directions for Time Series Anomaly Detection. In *International Conference on Learning Representations*. https://openreview.net/forum?id=H27kvyG4qf Released scores: KU Leuven RDR, V1, doi:10.48804/MCVI4N.